Abubakar Sadiq Shittu

Knoxville, Tennessee
ashittu@vols.utk.edu | 865-236-7456 | sadiq.info | linkedin.com/in/asshittu

RESEARCH STATEMENT

My research focuses on preventing malicious entities from compromising widely used software by advancing the adoption of cryptographic defenses like *commit signing* among software developers. Using empirical and human-centered methods, I investigate developer behaviors, tooling hurdles, and adoption patterns to identify why security tools succeed, are abandoned, or can be improved. Beyond software supply chain defenses, my broader goal is to identify generalizable principles that solve multi-decade grand challenges in usable key management across cryptographic systems.

EDUCATION

Ph.D. Computer Science 01/25–Present
University of Tennessee Knoxville, United States GPA: 3.67/4.00

- *Passed Comprehensive Examination:* 10/26
- *Expected Graduation:* 05/28
- *Advisor:* Dr. Scott Ruoti

M.Sc. Advanced Computer Science 09/23–09/24
Newcastle University, Newcastle upon Tyne, United Kingdom GPA: 4.00/4.00

- *Advisor:* Prof. Thomas Gross

B.Sc. Computer Science, First Class Honours 03/18–06/21
Fountain University Osogbo, Nigeria GPA: 4.63/5.00

- *Advisor:* Dr. Mohammed Lawal

PUBLICATIONS

[1] **Abubakar Sadiq Shittu**, Clay Shubert, John Sadik, and Scott Ruoti. "Setting Up TLS Authentication Was Hell": A Usability Study of Client Certificate Authentication. **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26), November 15–19, 2026, The Hague, Netherlands.* [**Acceptance Rate: TBD%**]. DOI: 10.1145/3830454.3846525. Preprint: arXiv:2604.14330.

[2] **Abubakar Sadiq Shittu**, John Sadik, and Scott Ruoti. "A Multi-Month Study of Git Commit Signing." **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26), November 15–19, 2026, The Hague, Netherlands.* [**Acceptance Rate: TBD%**]. DOI: 10.1145/3830454.3846532. Preprint: arXiv:2608.29283.

Manuscripts Under Review

[3] **Abubakar Sadiq Shittu**, John Sadik, Farzin Gholamrezae, and Scott Ruoti. "Analysis of Commit Signing on GitHub." Manuscript under review, 2026. Preprint: arXiv:2604.14014.

[4] John Sadik, **Abubakar Sadiq Shittu**, and Scott Ruoti. "A Worldwide Survey of Authentication Practices and Ecosystems".

[5] John Sadik, **Abubakar Sadiq Shittu**, and Scott Ruoti. "Lab study of new password generation interfaces".

EXPERIENCE

Graduate Research Assistant Knoxville, Tennessee, United States
USER Lab, University of Tennessee 01/25–Present

- Designed and conducted mixed-methods longitudinal user studies with 22+ participants on security systems to identify usability, functionality, and key-management barriers to adoption and correct use; analyzed qualitative self-reports, behavioral data from server logs, and standardized usability metrics (ASQ and SUS) using inferential statistics.
- Plan and conduct stakeholder interviews to examine factors influencing successful deployment of systems.
- Synthesize findings across multiple studies to identify generalizable design principles for future systems.
- Designed a stratified sampling measurement study of GitHub commit signing and built a Docker/Celery pipeline using REST and GraphQL APIs to asynchronously collect millions of user and commit records in PostgreSQL for analysis.

Information Technology Helpdesk Intern Nigeria
Nigerian National Petroleum Corporation 03/19–08/19

- Delivered on-demand IT support to 100+ users, diagnosing and resolving hardware and software issues daily.

AWARDS

- **Travel Grant**, *Summer School on Real-World Crypto and Privacy, Croatia* (2024).
- **Chevening Scholarship**, *UK Government-funded* (2023; approximately $70,000 in tuition, stipend, and travel support).

PEER REVIEW

Journals

- Journal of Cybersecurity 2025, 2026

Conferences

- 10th ACM Conference on Fairness, Accountability, and Transparency (FAccT) 2027

Artifact Evaluation Committees

- 33rd ACM Conference on Computer and Communications Security (CCS) 2026
- Network and Distributed System Security Symposium (NDSS '27) 2026

SKILLS

- **Programming:** Python, R, SQL, C/C++.
- **Analysis:** pandas, NumPy, scikit-learn, Matplotlib, Seaborn, ggplot2, dplyr.
- **Engineering:** PostgreSQL, Docker, Celery, Redis, ETL, API Integration.
- **Research:** Prolific, Qualtrics, NVivo, MAXQDA, LaTeX.
- **Tools:** Git, SonarQube, Wireshark, Linux/Unix, Windows.

AFFILIATIONS

- Graduate Student Member, **IEEE Computer Society** 2025 – Present
- Professional Member, **Association for Computing Machinery** 2026 – Present
- Volunteer, **Tennessee Extension**